move faster and migrate farther as compared to longer ones. The distance between DNA or RNA bands of a given length is determined by the percentage of agarose in the gel.

10.5 Centrifugation techniques

10.5.1 Principle

Centrifugation is a separation technique based on the properties of the particle in an applied centrifugal field. Particles which differ in size, shape and density sediment at different rates in a medium held in a tube. The rate of sedimentation depends upon the applied centrifugal field, density and radius of the particle as well as the density and viscosity of the suspending medium. The rate of sedimentation can be expressed as rpm (revolutions per minute) or g (gravitational force). The rate of sedimentation is dependent upon the centrifugal field (G) which is determined by the square of the angular velocity of the rotor (ω^2) and radial distance (r) from the axis of rotation, expressed in the equation$G = \omega^2 r$

The instrument used to hold the sample and generate the centrifugal force is called the centrifuge. This is accomplished through the use of various rotors of specific diameter. A rotor can hold the sample and spin along its own axis at different speeds to generate centrifugal force.

10.5.2 Types of Rotors:

Rotors are important components in a centrifuge and there are several types of rotors. The schematic representation of rotors used in centrifuge is shown in Fig.10.12.

1. **Fixed angle rotor:** In fixed angle rotor, particles move radially outward under the influence of a centrifugal field. The tubes are held at the angle 20 to 45 degrees. Particles have only a short distance to travel, to form a pellet at the bottom of the tube.

2. **Swinging bucket rotor:** The swinging bucket rotor, starts off in a vertical position and during acceleration of the rotor swing out to a horizontal position. During centrifugation, the solution in the tube, is aligned perpendicular to the axis of rotation and parallel to the applied centrifugal field, the tube returning to its original position as the rotor slows down.

3. **Vertical rotor:** Vertical rotor presents the shortest possible path for a particle. In this rotor, the pellet is deposited along the entire length of the tube.

4. **Zonal rotor:** The zonal rotors may be of the batch or continuous flow type. They are designed to minimize the wall effect that is encountered in swinging-bucket and fixed

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